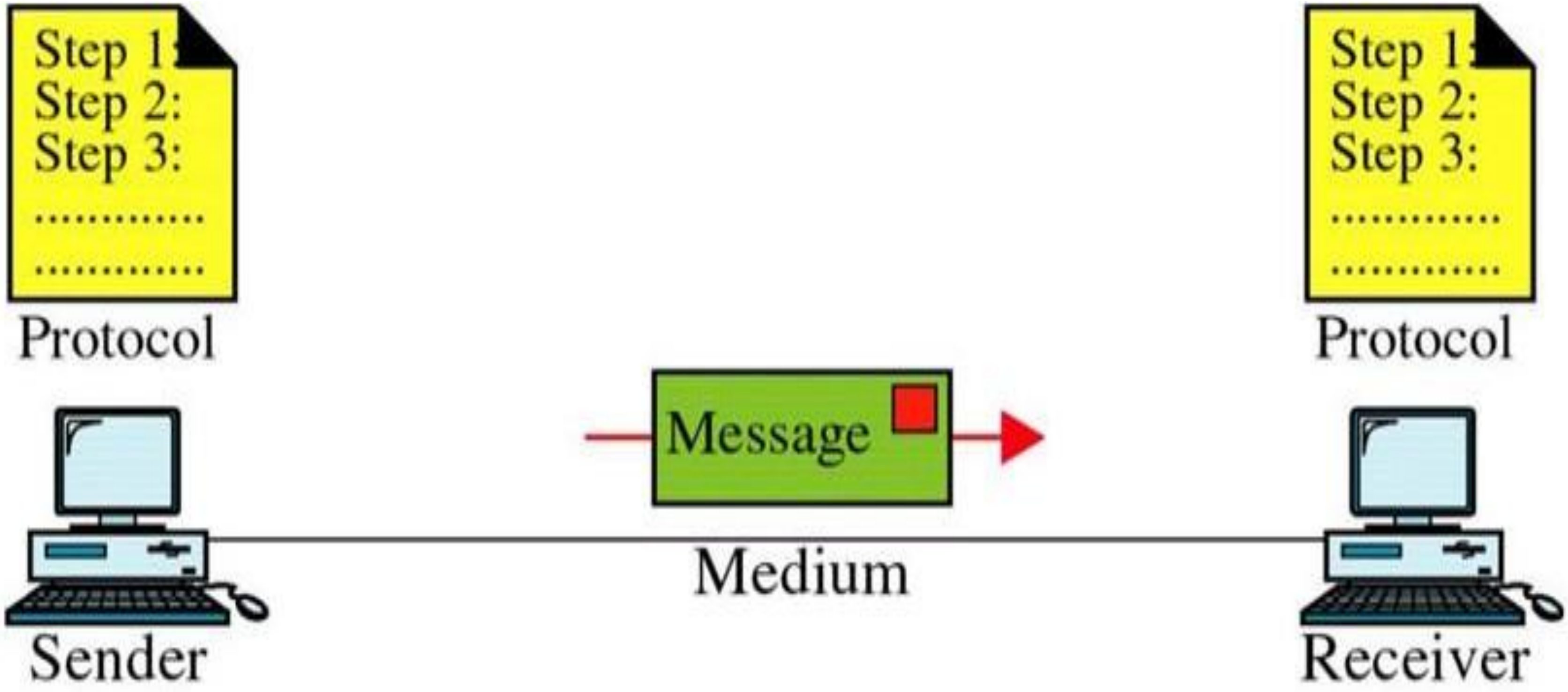
Unit 1

Fundamentals of Data Communication and Networking

1.1 Process of data communication and its components: Transmitter, Receiver, Medium, Message, Protocol.

- Data communications refers to the transmission of this digital data between two or more computers
- A computer network or data network is a telecommunications network that allows computers to exchange data.
- The physical connection between networked computing devices is established using either cable media or wireless media.
- The best-known computer network is the Internet.

Components of Data Communications System



Components of Data Communications System

- **A data communications system has five components**:

1. **Message**: The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender**: The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver**: The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. **Transmission medium**: The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
5. **Protocol**: A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

1.2 Protocols, Standards, Standard organizations, Bandwidth, Data Transmission Rate, Baud Rate and Bits per second.

- Protocol is a set of rules that govern all aspect of data communication between computers on a network.
- These rules include guidelines to regulate: access method, allowed physical topologies, types of cabling, and speed of data transfer.
- A protocol defines what, how, when it communicated.
- *Protocols are to computers*, *what language is to humans*. Since this article is in English, to understand it you must be able to read English. Similarly, for two devices on a network to successfully communicate, they must both understand the same protocols.

Elements of a Protocol

- The key elements of a protocol are **syntax, semantics and timing.**
- **Syntax:** (what is to be communicated?)
- The structure or format of the data.

For example, a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, and the rest of the stream to be the message itself

Elements of a Protocol

ii) Semantics:

(how it is to be communicated)

- Refers to **the meaning** of each section of bits.
- how is a particular pattern to be interpreted, and what action is to be taken based on that interpretation.

Eg. Does an address identify the route to be taken or the final destination of the message?

Elements of a Protocol

iii) Timing

Refers to two characteristics:

- When data to be sent
- How fast it can be sent

Eg. If a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and data will be largely lost.

sol:sender must send 1Mbps data and wait for ack before next 1Mbps of data.

Standards

- Standards are common rules followed by all devices so that they can work together properly.
- In simple words: Same rules for everyone.
- Example: All USB devices work because they follow the same standard.

Standards

- Data communication standards fall into two categories: ***de facto*** (meaning "by fact" or "by convention") and ***de jure*** (meaning "by law" or "by regulation").
- **De-facto**. Standards that have not been approved by an organized body but have been **adopted as standards through widespread use** are de facto standards. De facto standards are often established originally by manufacturers who seek to define the functionality of a new product or technology.
- **De-jure**. Those standards that have been legislated by an officially recognized body are de-jure standards.

Standards

- Standards are developed by cooperation among standards creation committees, forums, and government regulatory agencies.

Standards Creation Committees

1. International Standards Organization (ISO)
2. International Telecommunications Union (ITU)
3. American National Standards Institute (ANSI)
4. Institute of Electrical and Electronics Engineers (IEEE)
5. Electronic Industries Association (EIA)
6. Internet Engineering Task Force (IETF)

Bandwidth

- Bandwidth is also called as **data transfer rate**.
- Moving data from one point to another, in a given time period (usually a second), is called **bandwidth**.
- Bandwidth is indicated by **bits (of data) per second (bps).**

Bit Rate

- Bit Rate can be defined as the number of bit intervals per second.
- And bit interval is referred to as the time needed to transfer one single bit. In simpler words, the bit rate is the number of bits sent in one second, usually expressed in bits per second (bps). For example, kilobits per second (Kbps), Megabits per second (Mbps), Gigabits per second (Gbps), etc.

Baud rate

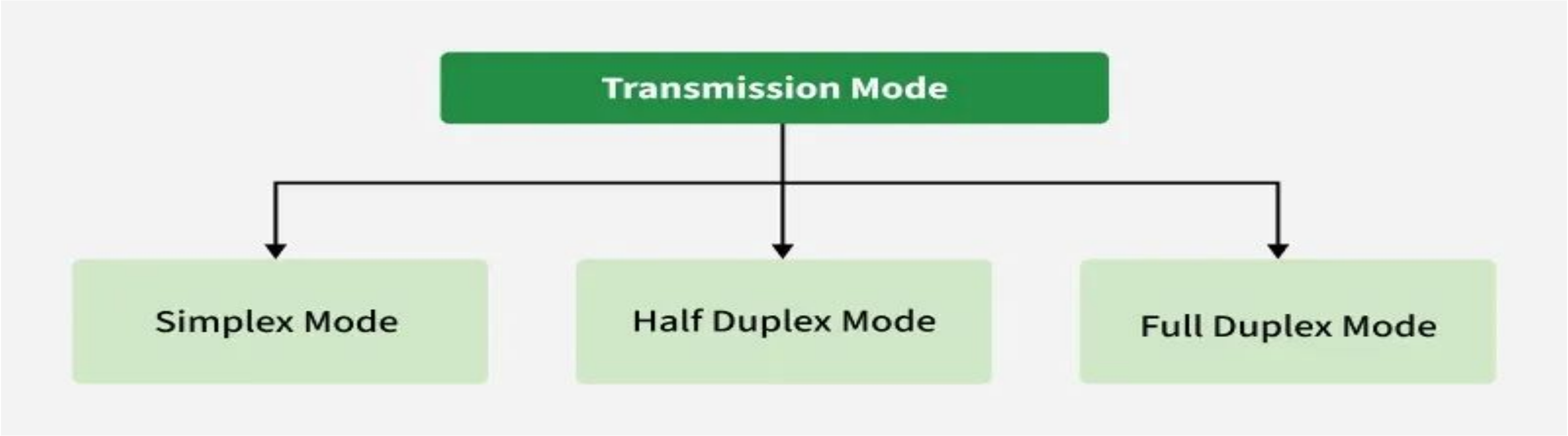
- Baud rate is defined as the number of signal units per second.
- It is always less than or equal to bit rate. It is represented as bauds or symbols/second.
- Baud rate is expressed in the number of times a signal can change on transmission line per second. Usually, the transmission line uses only two signal states, and make the baud rate equal to the number of bits per second that can be transferred.

Bit Rate Vs Baud Rate

BASIS FOR COMPARISON	BIT RATE	BAUD RATE
Basic	Bit rate is the count of bits per second.	Baud rate is the count of signal units per second.
Meaning	It determines the number of bits traveled per second.	It determines how many times the state of a signal is changing.
Term usually used	While the emphasis is on computer efficiency.	While data transmission over the channel is more concerned.
Bandwidth determination	Can not determine the bandwidth.	It can determine how much bandwidth is required to send the signal.

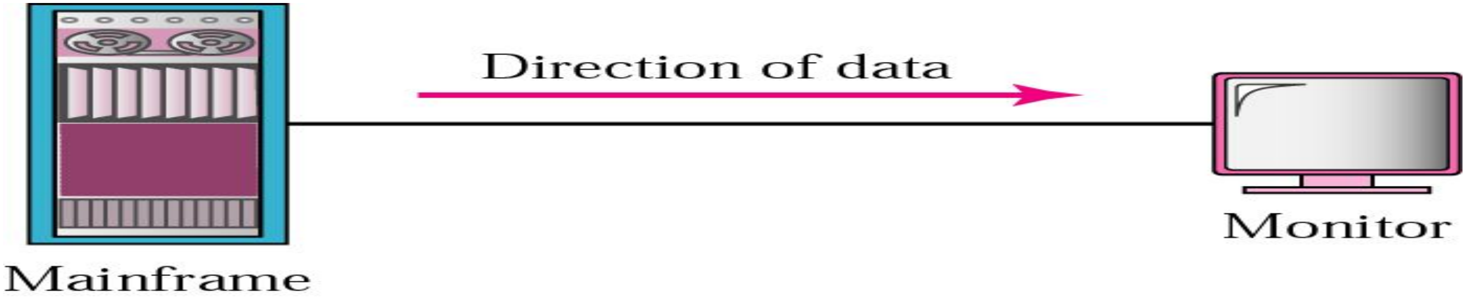
1.3 Modes of Communication

- Transmission Modes describe how data is exchanged between two devices over a communication channel.
- They specify whether information moves in a single direction or in both directions, and whether devices can transmit at the same time.



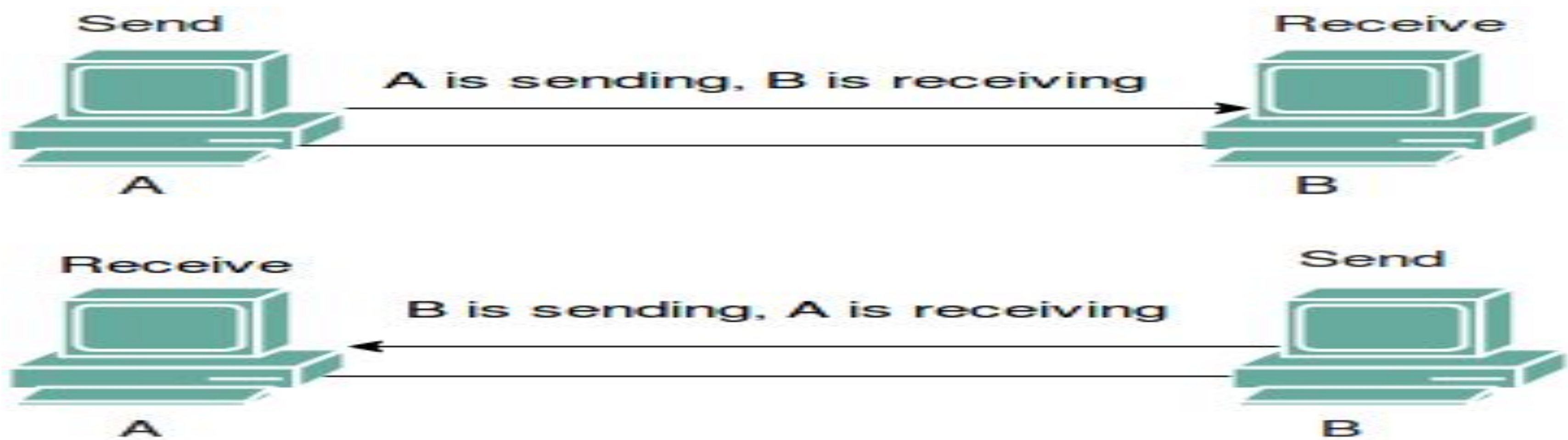
simplex mode

- In simplex mode, the communication is unidirectional.
- Only one of the devices on a link can transmit, the other can only receive. (Simplex Mode is a transmission mode in which communication occurs in only one direction, from a sender to a receiver, without any possibility of reverse data flow.)
- e.g. keyboards, monitors, etc.
- The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.
- No feedback or acknowledgment is supported.



Half-duplex mode

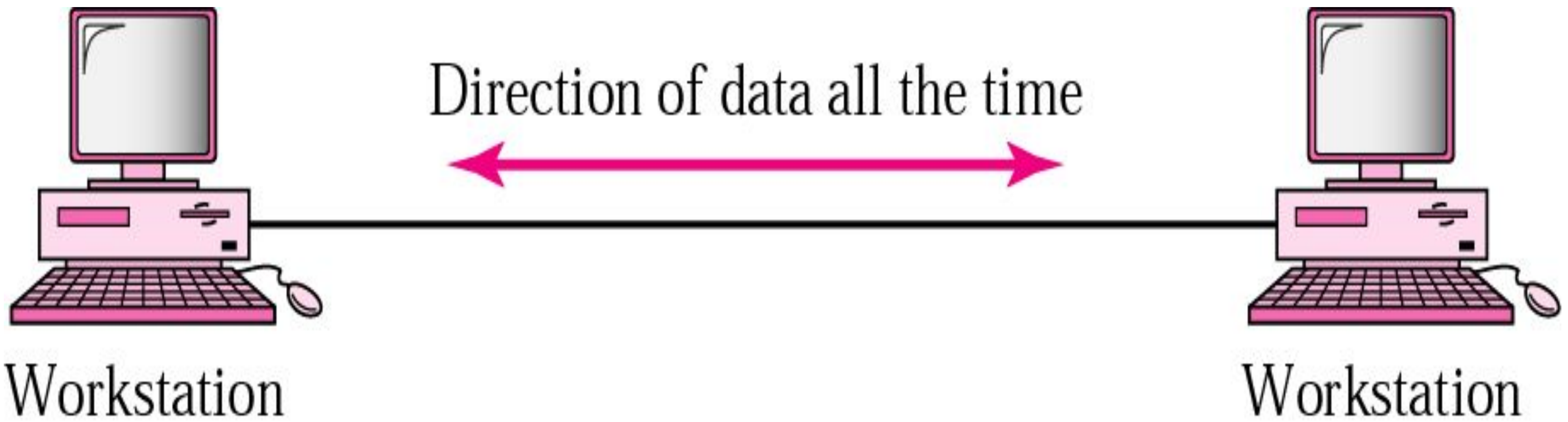
- In Half-duplex mode, each station can both transmit and receive, but not at the same time.



- When one device is sending, the other can only receive, and vice versa. The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait. In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.
- The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction .
- ***Example:*** *Walkie-talkie in which message is sent one at a time and messages are sent in both directions.*

Full duplex mode

- In full duplex mode, both stations can transmit and receive simultaneously.
- One common example of full duplex is the Telephone network.
- When two people are communicating by a telephone line, both can talk and listen at the same time.



1.4 Fundamentals of Computer Network

- ✔ DEFINITION
- ✔ NEED
- ✔ APPLICATION
- ✔ BENEFITS

❑ A computer network is a collection of interconnected devices that share resources and information. These devices can include computers, servers, printers, and other hardware.

Fundamentals of Computer Network

- Network: A group of connected computers and devices that can communicate and share data.
- Node: Any device that can send, receive, or forward data in a network. This includes laptops, mobiles, printers, earbuds, servers, etc.
- Networking Devices: Devices that manage and support networking functions. This includes routers, switches, hubs, and access points.
- Transmission Media: The physical or wireless medium through which data travels between devices.
- Wired media: Ethernet cables, optical fibre.
- Wireless media: Wi-Fi, Bluetooth, infrared
- Service Provider Networks: This includes internet providers, mobile carriers, etc.

APPLICATIONS

1. Banking
2. Video conferencing
3. Marketing
4. School
5. Radio
6. Television
7. E-mail
8. Companies

Application of Network

- Business applications: Computer networks are often used by businesses to ensure impact communication, to share resources, and to allow their employees to access the whole system and applications from remote locations.
- Educational applications: Online networks are widely employed in educational institutions allowing students to access educational possibilities, share knowledge, and collaborate with their professors.
- Healthcare applications: The healthcare sector has benefited a lot from the computer networks, which are used to store and share patient details thus allowing healthcare providers to provide more personalized treatment.
- Entertainment applications: Besides that with computer networks, you can entertain yourself with online games, streaming movies and music, or utilization of social media.
- Military applications: Military networks are often closed and not used for general communication, which ensures the safety of military information.
- Scientific applications: Scientific research heavily depends on computer networks because they will help establish collaboration among researchers and facilitate the sharing of data and information.

Benifits of Computer Network

- Sharing of Information
- Inexpensive set-up: Shared resources=> means reduction in hardware costs.
- Resource sharing
- Flexible Handling.
- Facilitating Centralized Management.
- E-mail Services
- Backing up data.

Advantages/Benifits

1. **File sharing**: **Computer network is that it allows file sharing** and **remote file access.**

A person sitting at one workstation that is connected to a network can easily see files present on another workstation, provided he/she is authorized to do so.

2. Resource Sharing:

A computer network provides a **cheaper alternative by the provision of resource sharing**. All the computers can be interconnected using a network and just one **modem & printer can efficiently provide the services to all users.**

Advantages/Benifits

3. Inexpensive set-up:

    Shared resources=> means reduction in **hardware costs**. Shared **files=>** reduction in **memory** requirement=>reduction in file **storage** expenses.

4. Flexible Handling: log on from anywhere and access his/her files. This offers flexibility to the user

5. Centralized Management- management of r**esources in the organization, centrally through client server architecture**.

6. Backing up data: Creating **backup files** and **restoring** them becomes much **easier.**

7. E-mail Services

File sharing

- File sharing is the **primary feature** of network.
- File sharing requires a shared directory or disk drive to each) which many users can access over the network.
- When **many users are accessing the same file on the network**, more than one person can make changes to a file at the same time. They might both making conflicting changes simultaneously.

E- mail services:

- E-mail is extremely valuable & important feature for communication within organization or outside the people in world.

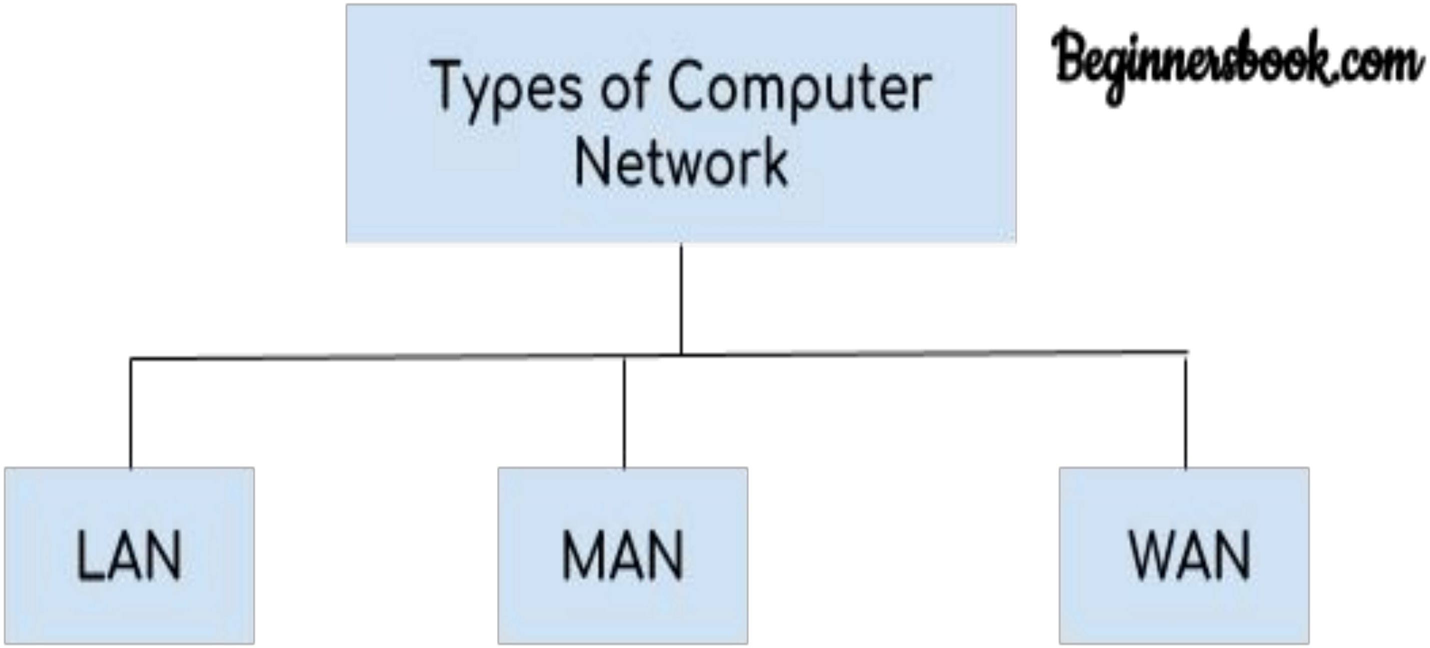
Remote access

- Using this feature user can access their file & e-mail, when they are travelling or working on remote location.
- It enables users to access to centralized application, stored private or shared files on LAN.

Internet & Intranet

- **Internet**: It is public network. This consists of thousands of individual networks & millions of computers located around the world.
- Internets have many different types of services available such as e-mail, the web & Usenet newsgroups.
- **Intranet**: It is private network or it is company's own network. Company use this feature for internal use.
  - For example: company establish its own web server, for placing documents such as employee handbooks, purchases form or other information that company publishes for internal use. It also has internet services such as FTP servers or Usenet servers.

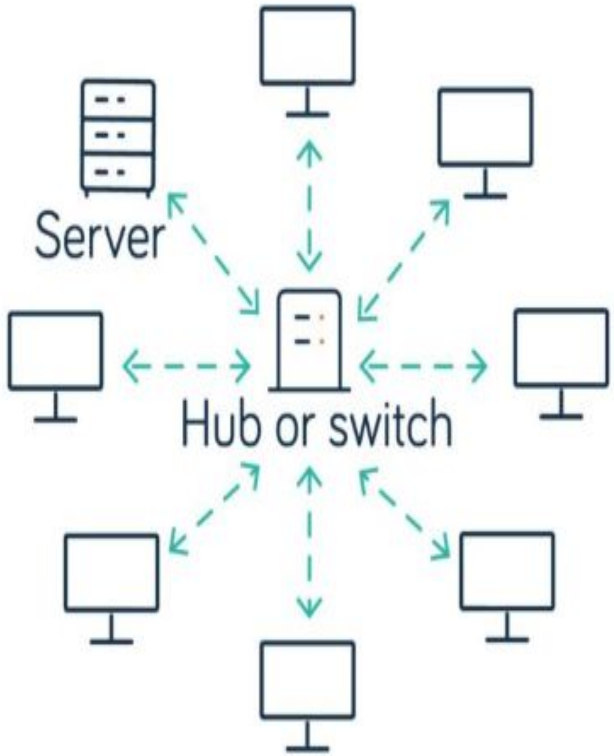
Classification of Network

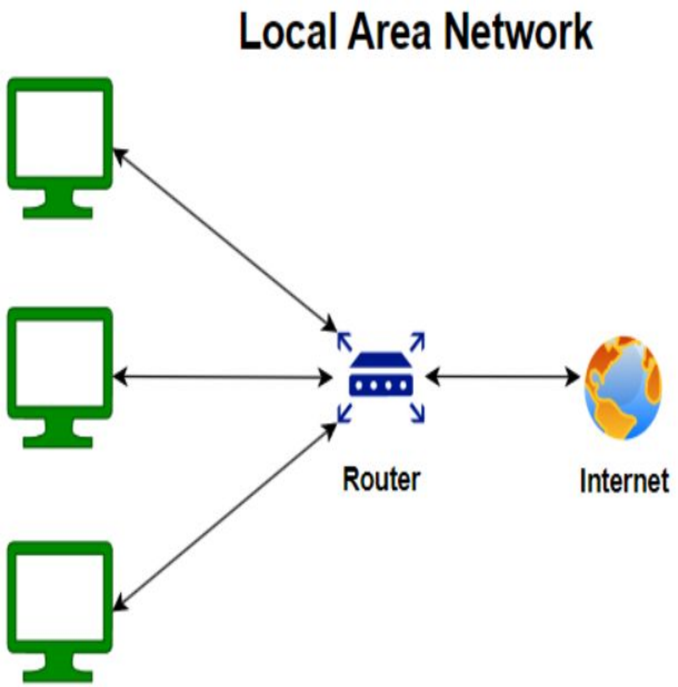


LAN - Local Arean Network

- Local area network is a group of computers connected with each other in a small places such as school, hospital, apartment etc.
- LAN is secure because there is no outside connection with the local area network thus the data which is shared is safe on the local area network and can’t be accessed outside.
- LAN due to their small size are considerably faster, their speed can range anywhere from 100 to 100Mbps.

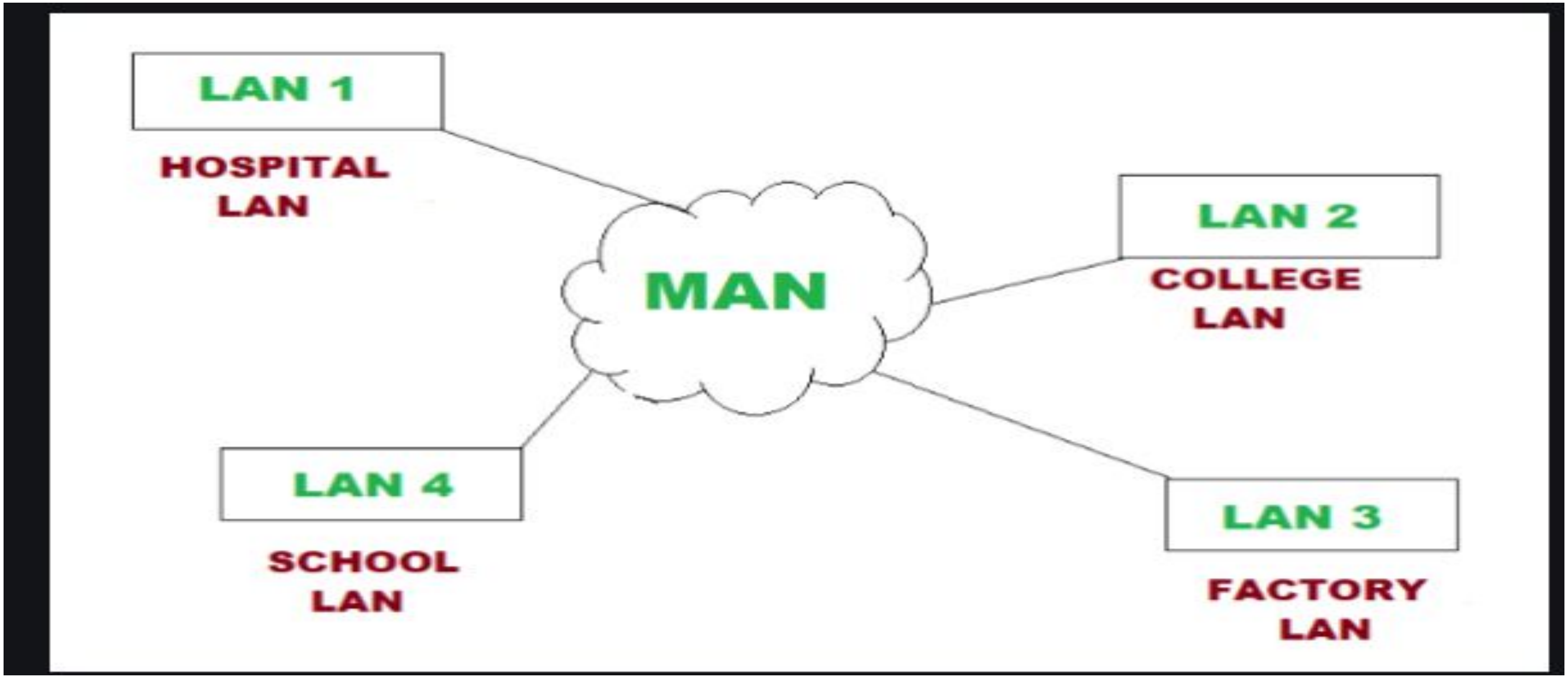
LAN - Local Arean Network





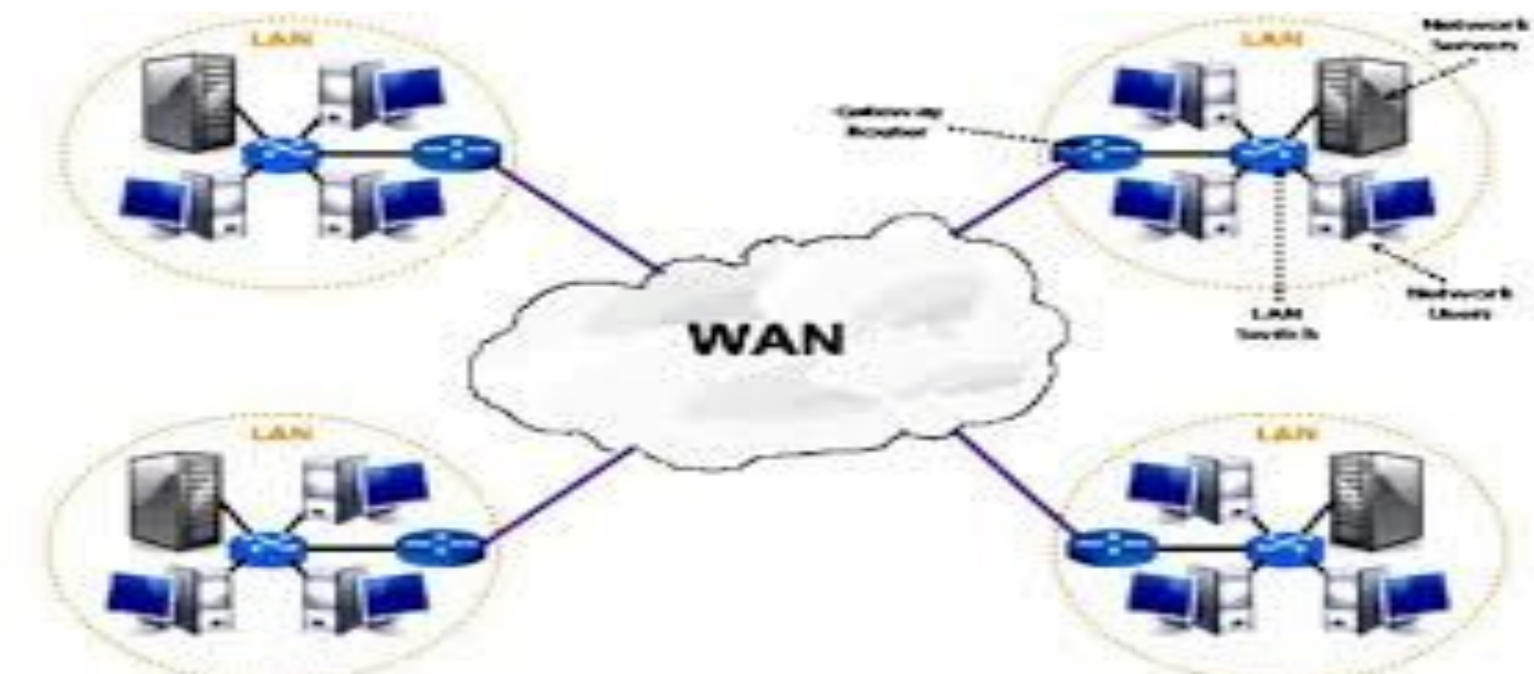
MAN - Metropolitan Area Network

- Metropolitan Area Network (MAN) connects multiple Local Area Networks (LANs) across a city or large campus, acting as an intermediate network between smaller LANs



WAN - Wide Area Network

- Wide area network provides long distance transmission of data. The size of the WAN is larger than LAN and MAN. A WAN can cover country, continent or even a whole world. Internet connection is an example of WAN. Other examples of WAN are mobile broadband connections such as 3G, 4G etc.



WAN - Wide Area Network

- **Coverage Range:** Country, continent, or global
- **Key Technologies:** Leased lines, Satellite links, Internet
- **Real-World Example:** The Internet
- **When It Is Used:** To enable long-distance and global communication

LAN vs MAN vs WAN

Parameter	LAN	MAN	WAN
Full Form	LAN is an acronym for Local Area Network.	MAN is an acronym for Metropolitan Area Network.	WAN is an acronym for Wide Area Network.
Definition and Meaning	LAN is a network that usually connects a small group of computers in a given geographical area.	MAN is a comparatively wider network that covers large regions- like towns, cities, etc.	The WAN network spans to an even larger locality. It has the capacity to connect various countries together. For example, the Internet is a WAN.
Network Ownership	The LAN is private. Hospitals, homes, schools, offices, etc., may own it.	The MAN can be both private or public. Many organizations and telecom operators may own	The WAN can also be both private or public.
Maintenance and Designing	Very easy to design and maintain.	Comparatively difficult to design and maintain.	Very difficult to design and maintain.
Speed	LAN offers a very high Internet speed.	MAN offers a moderate Internet speed.	WAN offers a low Internet speed.
Delay in Propagation	It faces a very short propagation delay.	It faces a moderate propagation delay.	It faces a high propagation delay.
Uses	Schools, homes, colleges, hospitals, offices, etc., can privately use it.	It basically covers a city, a small town, or any given area with a bigger radius than the LAN.	It covers an entire country, a subcontinent, or an equivalent area.

1) Analyze the following situations and classify each as a LAN, MAN, or WAN. Justify your classification by explaining the geographical coverage and typical use of the network.

- A university computer lab connecting all computers within the lab.
- A city-wide cable TV network providing services across a metropolitan area.
- The Internet connecting users and networks around the world.

Ans

Situation	Network Type	Justification
University computer lab	LAN	Connects devices within a limited area such as a room, building, or campus.
City-wide cable TV network	MAN	Covers a city or metropolitan area and connects users across the city.
Internet	WAN	Connects networks across countries and continents, covering a very large geographical area.

1.6 Basics of Cloud Networking and IoT Communication

- ✔ Overview of IoT, cloud networking, devices, sensors, and applications.
- ❑ Cloud Networking-

A networking approach where network resources and services are hosted and managed through cloud infrastructure.

Examples: AWS, Azure, Google Cloud

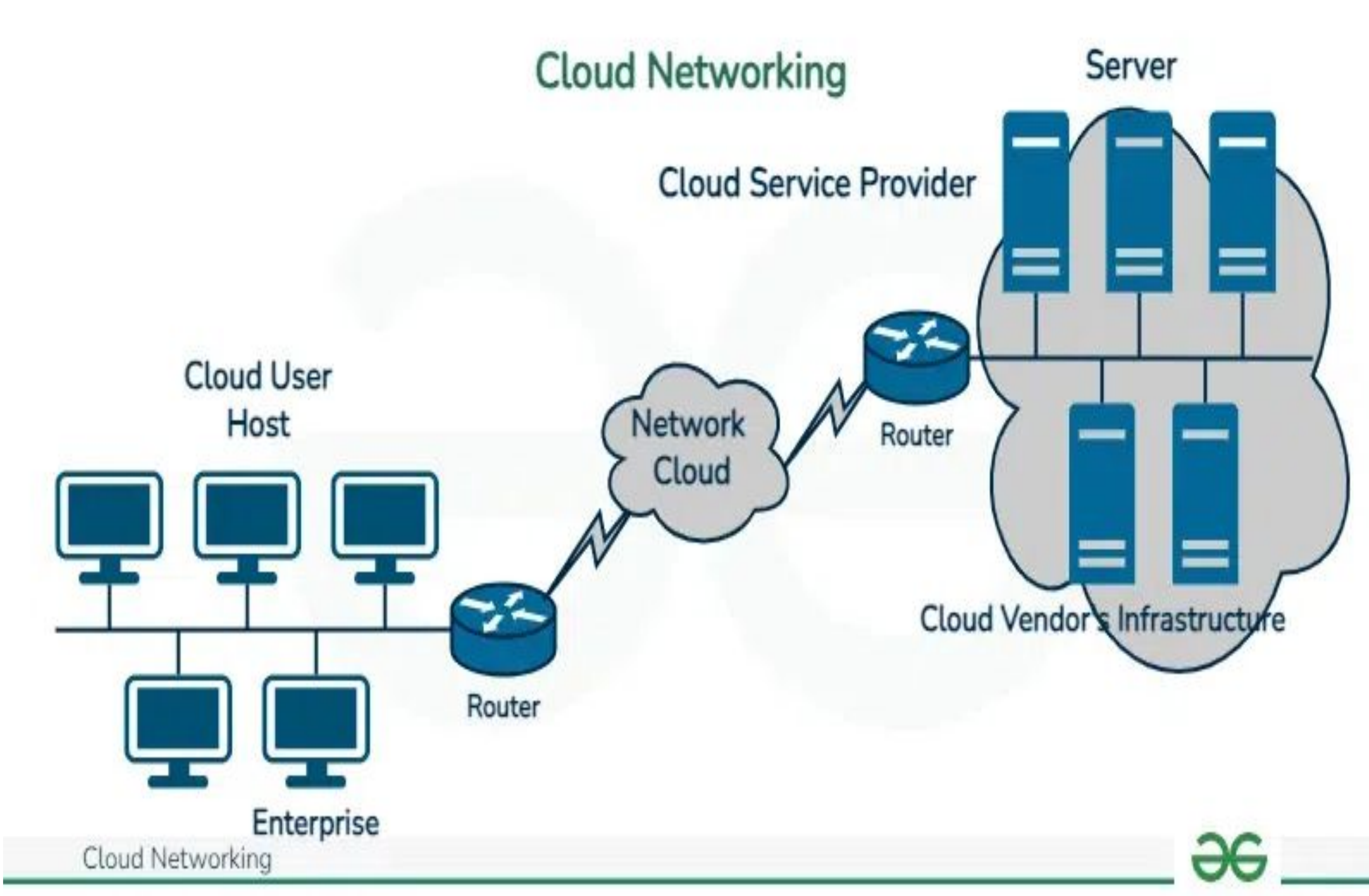
- ❑ What is IoT?
- ❑ IoT (Internet of Things) connects physical devices to the internet
- ❑ Enables data collection and remote control
- ❑ Examples: Smart homes, wearables, industrial automation

"IoT and cloud networking together enable smart systems"

Cloud Networking

- Cloud networking is an IT infrastructure where some or all of an organization's networking capabilities are hosted in a cloud environment. Instead of relying entirely on physical hardware, organizations use virtualized tools like virtual routers, firewalls, and load balancers to connect users, applications, and data.

Cloud Networking



Cloud Networking

1. Cloud User (Enterprise)

- This is the client side, representing a local corporate office, institution, or end-user environment.
- **Hosts**: The local client devices (desktop computers, laptops, or workstations) used by employees to request data, apps, or services.
- **Local Network Bus**: The black horizontal lines represent a local area network (LAN) topology that links all the host computers together inside the office.
- **Router**: A local hardware or software gatekeeper that directs data packets out of the internal enterprise network and into the wider internet.

2. Network Cloud

- This sits directly in the center, acting as the bridge between the user and the vendor.
- **The Internet / WAN**: It represents the public internet or a private Wide Area Network (WAN).
- **Data Transit**: When a user clicks a button, data travels from the enterprise router, through this digital cloud network, and straight to the provider.

Cloud Networking

3. Cloud Service Provider (Cloud Vendor's Infrastructure)

- This represents the remote data center owned by a vendor (such as Amazon Web Services, Microsoft Azure, or Google Cloud Platform).
- **Router**: A high-capacity border router that receives incoming user traffic from the internet cloud and passes it safely into the data center.
- **Servers**: Massive racks of powerful physical computers that store websites, run application software, process computational workloads, and host database systems.

Features & Application of Cloud Networking

Feature of cloud Networking:

- On-demand access to network resources
- Scalability and flexibility
- Remote accessibility
- Cost-effective management
- High availability and reliability

Application of Cloud Networking

- Online storage
- Video conferencing
- Web hosting
- Cloud-based applications

IoT
1. Devices

- Hardware components that collect and transmit data
- Examples: Smart thermostats, cameras, wearable devices

2.Sensors in IoT

- Sensors are devices that detect physical conditions and convert them into digital data.
- Examples: Temperature, humidity, motion, light

Sensor	Purpose
Temperature Sensor	Measures temperature
Humidity Sensor	Measures moisture in air
Motion Sensor	Detects movement
Light Sensor	Measures light intensity
Gas Sensor	Detects gases and pollutants
Pressure Sensor	Measures pressure
Proximity Sensor	Detects nearby objects

Roles of sensors in IoT

- **Data Collection**
  - Sensors continuously measure real-world parameters such as temperature, humidity, pressure, light, motion, sound, and air quality.
  - They provide the raw data that IoT systems use.
- **Real-Time Monitoring**
  - Sensors enable continuous observation of environments, equipment, or processes.
  - Example: Monitoring the temperature inside a cold storage facility.
- **Automation** Sensor data can trigger automatic actions without human intervention.
  - Example: A motion sensor turns on lights when someone enters a room.
- **Decision Making** The collected data is analyzed to identify patterns, detect abnormalities, or predict future events.
  - Example: Vibration sensors on machines help predict equipment failures before they occur.
- **Remote Access and Control** Sensors send data over the internet, allowing users to monitor and manage devices from anywhere using smartphones or computers.
- **Improving Efficiency** Sensors help optimize resource usage, reduce waste, and lower operational costs.
  - Example: Soil moisture sensors help automate irrigation, reducing unnecessary water use.

Roles of sensors in IoT

- **Example**

In a smart home:

- A temperature sensor detects room temperature.
- The data is sent to a smart thermostat.
- If the temperature rises above a set limit, the thermostat automatically switches on the air conditioner.

How IoT Communication Works

- IoT communication generally follows this flow:

Sensor → IoT Device → Network → Cloud → Application → User/Actuator

How IoT Communication Works

1. **Sensors collect data** (e.g., temperature, motion).
2. **IoT devices process and send data**.
3. Data is transmitted through a **communication network**.
4. **Cloud systems store and analyze data**.
5. **Applications display results**
6. **Actuators perform actions** based on commands.

❖ In a smart home:

- A motion sensor detects movement.
- Data is sent to a smart hub.
- Cloud processes the data.
- Lights automatically turn ON.
- User can also control lights from a mobile app.

Applications of IoT

- **Smart Homes**
  - Smart lighting
  - Smart locks
  - Security cameras
- **Healthcare**
  - Remote patient monitoring
  - Fitness trackers
- **Agriculture**
  - Soil moisture monitoring
  - Smart irrigation
- **Smart Cities**
  - Traffic management
  - Smart parking
  - Waste management
- **Industrial IoT (IIoT)**
  - Predictive maintenance
  - Factory automation

References

- 1.3 https://www.geeksforgeeks.org/computer-networks/transmission-modes-computer-networks/

Cloud Networking Core Components

- **Virtual Private Cloud (VPC)**: Isolated, custom network environments built within a public cloud.
- **Load Balancers**: Managed services that distribute incoming web traffic across multiple servers to prevent downtime.
- **Virtual Routers & Firewalls**: Digital routing mechanisms and security barriers used to control data packets and block traffic.
- **Content Delivery Networks (CDNs)**: Networks of localized cache servers that speed up data delivery to end-users